Practice workbook · Explain how enzymes work in biosensors and test strips

Glucose strips and sensors using immobilised enzymes behind a partially permeable membrane.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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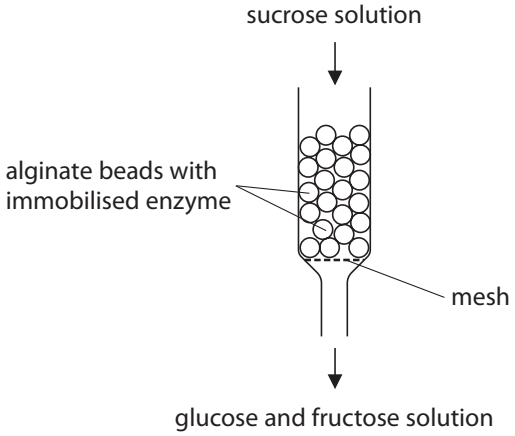
Type ID: HBI-ENZYME-BIOSENSOR-FUNCTION

1. Question 6 · part b.ii · 4 marks

November 2020 Paper 1

Worked-example retry · Attempt part b.ii only.

6 (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.



(i) Name the immobilised enzyme used in this process.

(1)

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(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

(3)

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(iii) State two other advantages of using immobilised enzymes.

(2)

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2 .....

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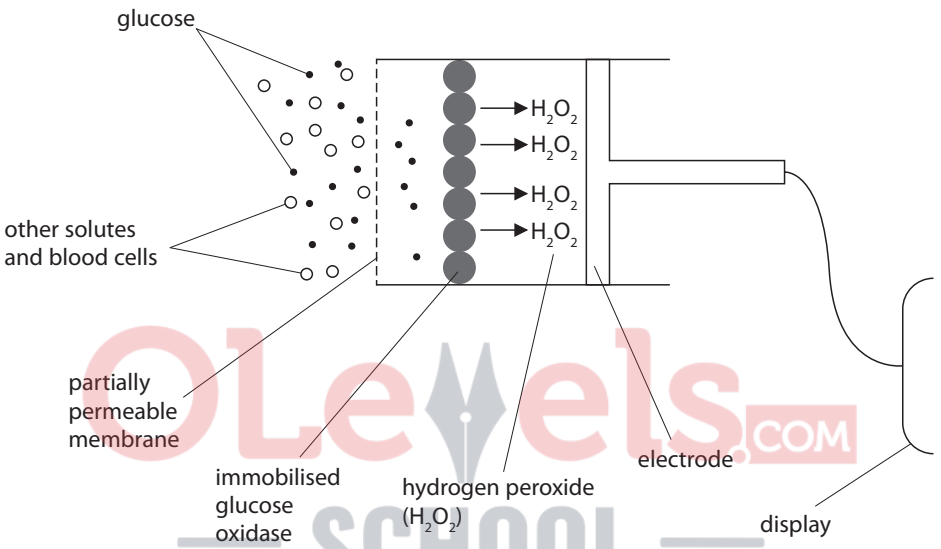
1. Question 6 · part b.ii · 4 marks

November 2020 Paper 1

Worked-example retry · Attempt part b.ii only.

(b) Biosensors that contain immobilised enzymes can be used to measure the levels of glucose in blood.

The diagram shows this type of biosensor.



(i) Explain the function of the partially permeable membrane.

(2)

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1. Question 6 · part b.ii · 4 marks

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Worked-example retry · Attempt part b.ii only.

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(ii) Explain the function of the immobilised enzyme in the biosensor.

Use information from the diagram to help your answer.

(4)

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(Total for Question 6 = 12 marks)

1. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

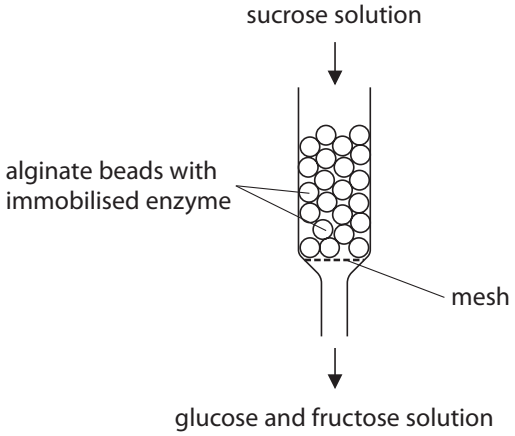
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2. Question 6 · part b.i · 2 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

6 (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.



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(1)

.....

(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

(3)

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(iii) State two other advantages of using immobilised enzymes.

(2)

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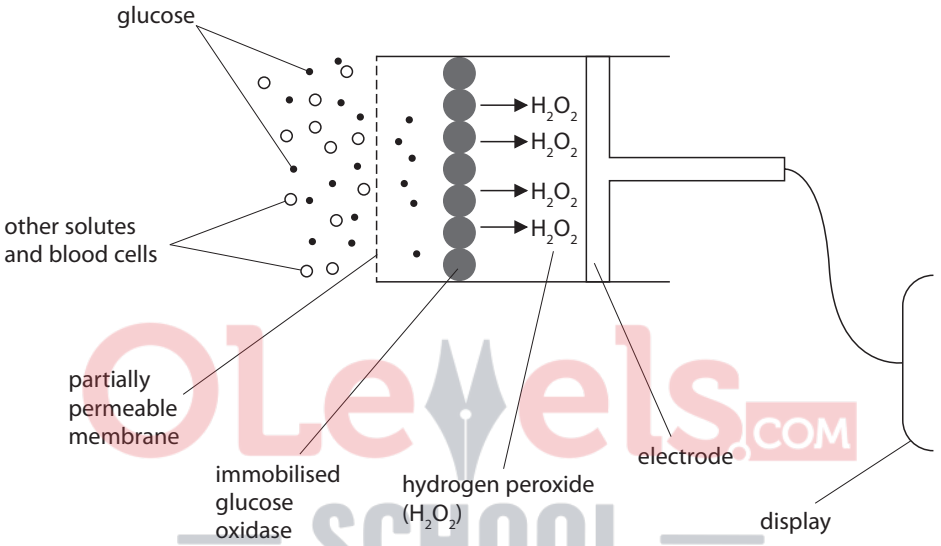
2. Question 6 · part b.i · 2 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

(b) Biosensors that contain immobilised enzymes can be used to measure the levels of glucose in blood.

The diagram shows this type of biosensor.



(i) Explain the function of the partially permeable membrane.

(2)

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2. Question 6 · part b.i · 2 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

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(ii) Explain the function of the immobilised enzyme in the biosensor.

Use information from the diagram to help your answer.

(4)

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(Total for Question 6 = 12 marks)

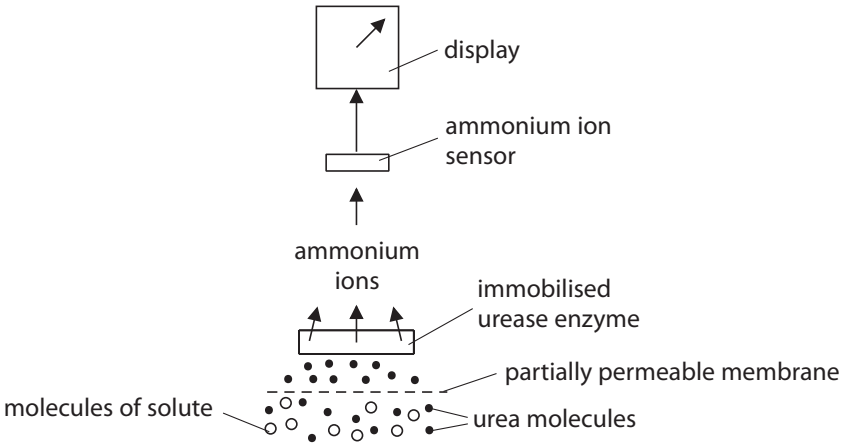
Working and answer

3. Question 3 · part a.ii · 2 marks

June 2021 Paper 2

Further practice · Attempt part a.ii only.

3 The diagram shows a biosensor that can be used to measure the concentration of urea in urine.



(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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3. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

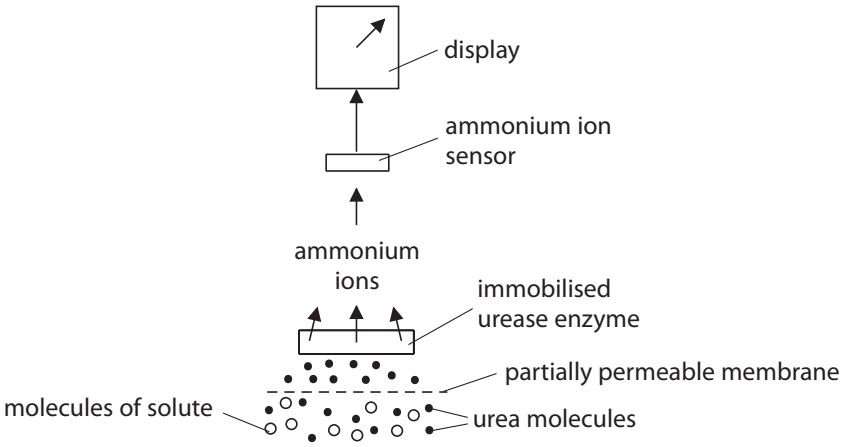
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4. Question 3 · part a.iii · 3 marks

June 2021 Paper 2

Further practice · Attempt part a.iii only.

3 The diagram shows a biosensor that can be used to measure the concentration of urea in urine.



(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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4. Question 3 · part a.iii · 3 marks

June 2021 Paper 2

Further practice · Attempt part a.iii only.

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(b) A scientist wants to use the biosensor to compare the urea concentration of two urine samples.

Explain why it is important to do the tests at the same temperature.

(4)

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(c) Immobilised enzymes are also used in industry to produce useful molecules.

Give two advantages of using immobilised enzymes instead of enzymes that are free to move in solution.

(2)

1

2

(Total for Question 3 = 12 marks)

4. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

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5. Question 7 · part b · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b only.

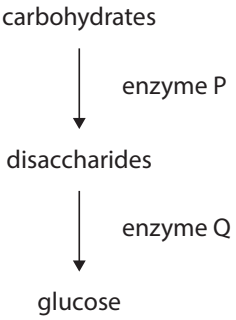
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7 (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.



Name enzyme P.

(1)

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(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

(2)

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(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

(1)

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5. Question 7 · part b · 2 marks · Working and answer

Working and answer

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